



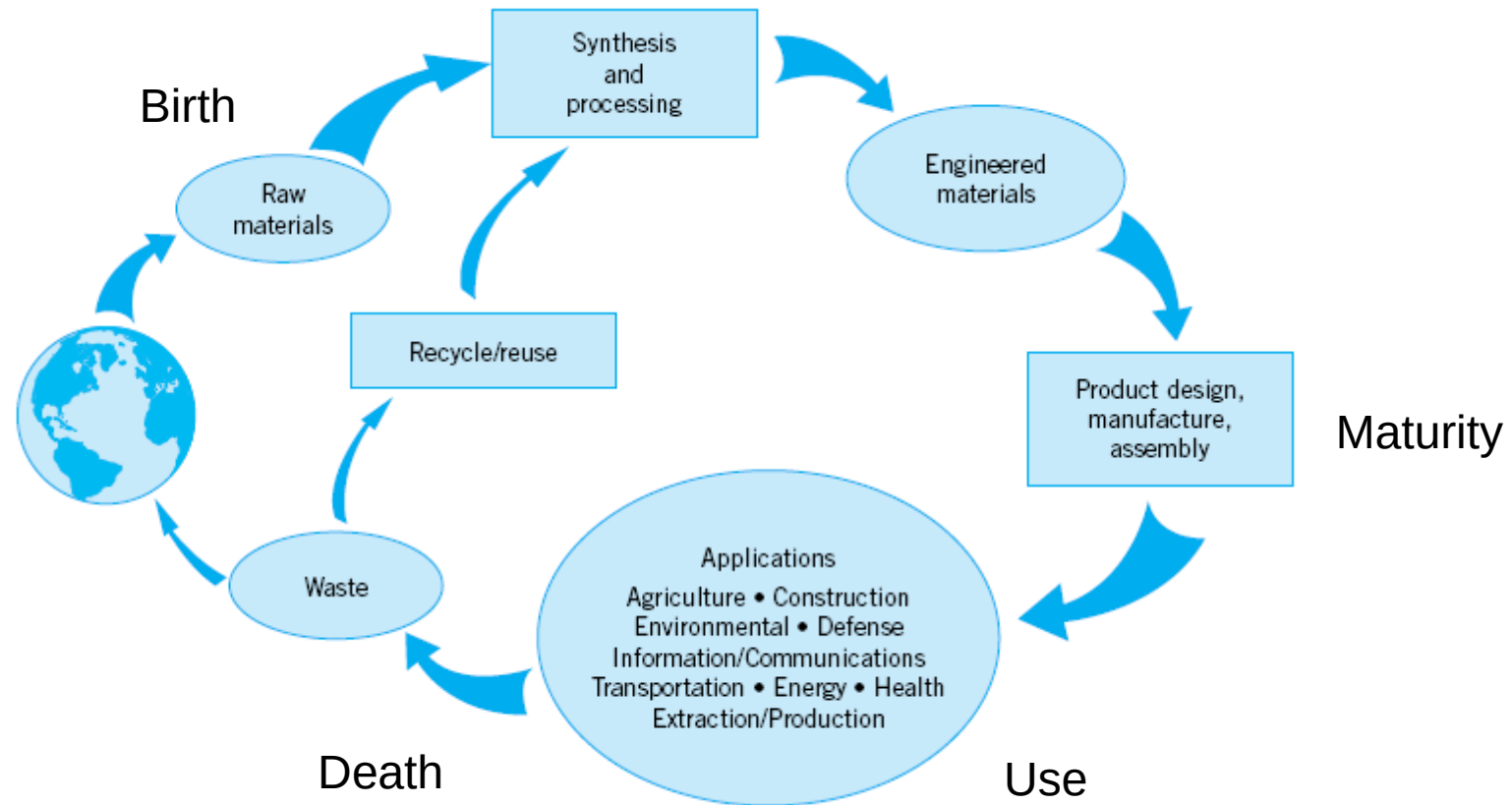
Life Cycle Assessment

ENGEN370 – Mark Lay



Product life cycle

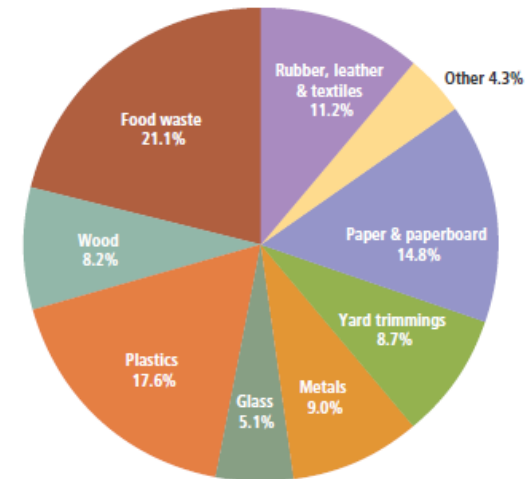
- 15 billion tonnes of raw material extracted from earth each year



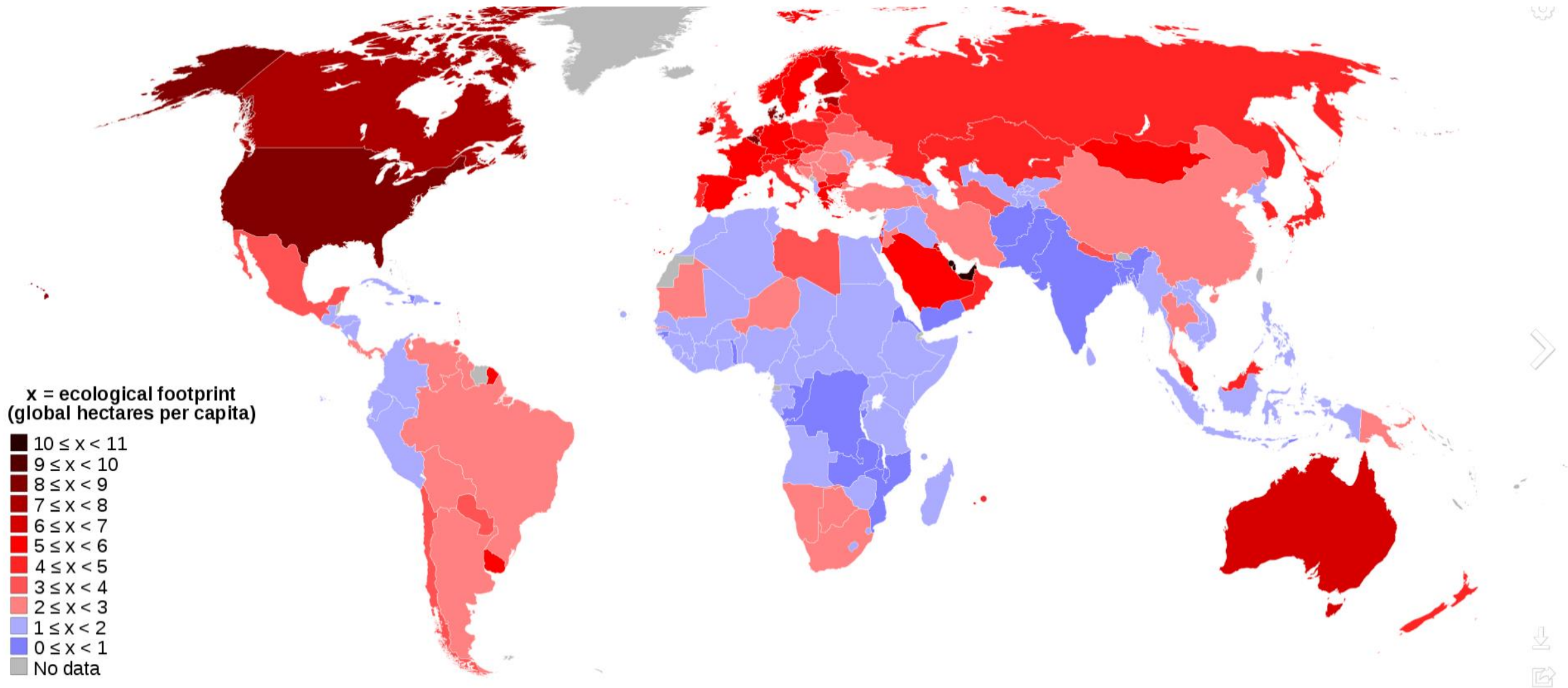
Municipal solid waste

- US – 250 million tonnes per year
 - 65 million tonnes recycled
 - 21 million tonnes composted
 - 29 million tonnes combusted
- NZ – 2.5 million tonnes per year
 - 75% could be potentially recycled, recovered, reused
 - 2008 Waste Minimization Act

Figure 7. Total MSW Discards (by material), 2012
164 Million Tons (after recycling and composting)

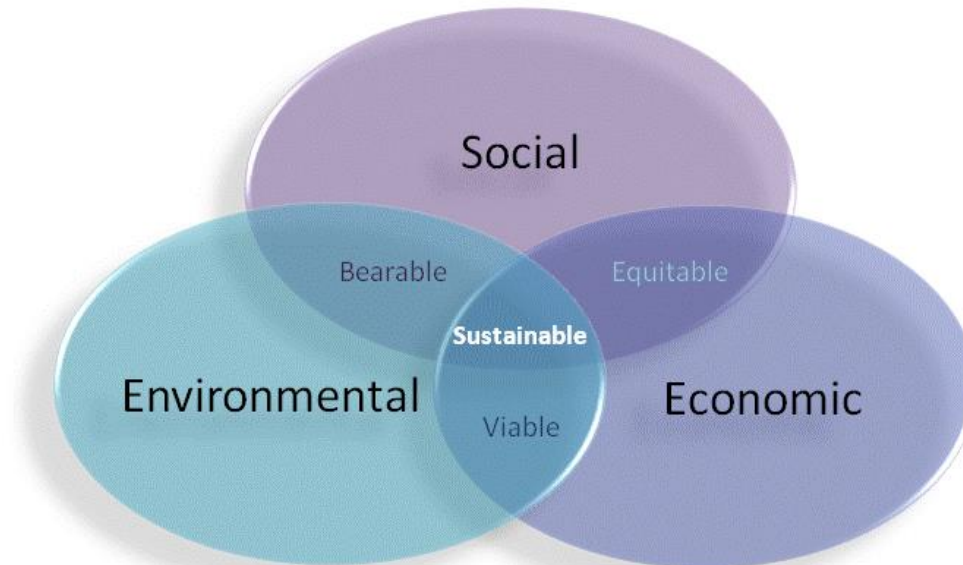


Ecological Footprint – 1.7x Earth



Sustainable Development

- Preserve genetic diversity
- Maintain agricultural, marine and forest systems
- Sustainable use of renewable resources
- Avoid degrading physical resources





- Set in 2015
- 193 countries

ISO 14040 - LCA

- Technique for assessing environmental and potential impacts associated with a product or service
 - Compiling an inventory of inputs and outputs of a product system
 - Evaluating impacts of those inputs and outputs
 - E.g. Energy demand, GWP, CO₂, CO, VOC, NO_x, SO_x, PM, photochemical oxidant potential, acidification, eutrophication, toxicity
 - Interpreting the results in relation to the objectives of the study
- Cradle to grave; cradle to gate; gate to gate; gate to grave
 - Cradle – birth of a material
 - Grave – death of a product and disposal
 - Gate – a stage in the production of a product

LCA

- Why is it useful?
 - Identify opportunities to reduce environmental impact – e.g. the tea bag
 - Decision making in industry, government or non-governmental organisations
 - Selection of relevant indicators of environmental performance
 - Marketing
- One of a set of environmental management techniques
 - Risk assessment
 - Environmental performance evaluation
 - Environmental auditing
 - Environmental impact assessment

Humble Tea Bag



LCA

- Can routinely use to improve products to make them more sustainable
 - Phillips
 - Unilever
 - Proctor and Gamble



LCA

- Equivalents
 - Mass and energy and whole life cost accounting for manufacturing processes
 - Cost accounting for different management/governmental strategies
 - E.g. whole life cost of weapons systems including use, maintenance and disposal
 - Cost benefit analysis for maximizing welfare – 1971 US executive Order 20503 – Quality of Life

First known LCA

- Coca Cola – 1969
 - Environmental impact of packaging
 - Resource use and disposal
 - Recyclable materials – steel, aluminium, glass and plastic (PET)
 - 2000 ultraglass contour bottle
 - 40% stronger, 20% lighter, 10% less expensive
 - Eliminated 52,000 tons of glass
 - CO₂ reduction of 26,000 tons per year
 - Reducing packaging weight



First links to public policy

- Netherlands – 1980s
 - Department of Environmental Management led by Peter Winsemius
 - Looked at water, air, soil
 - Acidification, eutrophication, diffusion of toxic substances, waste disposal, noise, odour, air pollution
 - Dutch companies Unilever and Philips routinely use it

Standardisation of LCA

- 1990's – Society of Environmental Toxicology and Chemistry (SETAC)
 - Brought LCA practitioners together
 - Harmonized methods, frameworks, practices
- 1994 - ISO – formalized standards, been ongoing since

LCA

- Limitations
 - Does not address economic or social impacts
 - Choices and assumptions made in carrying out the LCA are subjective
 - E.g. economic vs mass allocation, allocation of impacts to wastes?
 - Depth and detail will vary depending on goal and scope of the LCA
 - Models used are limited by their assumptions
 - Steady state, but what about Bhopal, Chernobyl, Fukushima?
 - LCA results may not be applicable for other settings
 - NZ (80% renewable) vs Aus (14.6% renewable) electricity
 - Accuracy of results dependent on quality, accessibility of data
 - Data not available, so have to make assumptions or find other sources to estimate
 - Temporal or spatial impacts not accounted for
 - Point source emissions, midnight valve, sudden spikes in emissions e.g. fire

ISO 14040

- ISO – International Organisation for Standardisation
- Membership of 161 national standards bodies
- Collaborates with 700 international, regional and national organisations
- Financed by subscriptions
- Based in Geneva, Switzerland
- Published 22136 International Standards
- Standards New Zealand is the NZ member for ISO

ISO 14000 family

- ISO 14040 – Life cycle assessment – Principles and framework
- ISO 14044 - Life cycle assessment – Requirements and guidelines
- ISO 14045 - Eco-efficiency assessment of product systems - Principles, requirements and guidelines
- ISO 14046 - Water footprint - Principles, requirements and guidelines
- ISO 14047 - Life cycle assessment - Illustrative examples on how to apply ISO 14044 to impact assessment situations

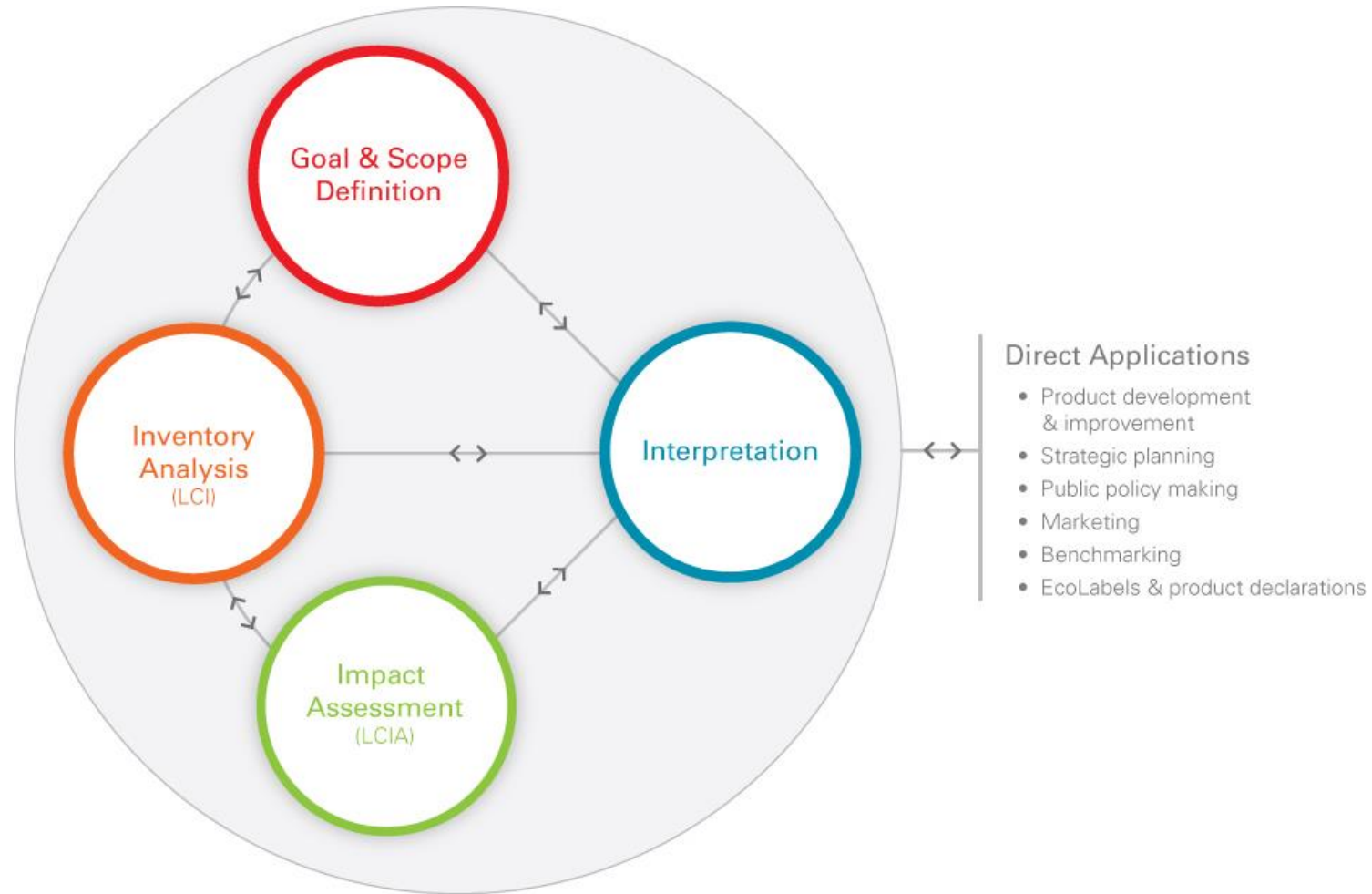
ISO 14000 family

- ISO 14048 - Life cycle assessment - Data documentation format
- ISO 14049 - Life cycle assessment - Illustrative examples on how to apply ISO 14044 to goal and scope definition and inventory analysis
- ISO 14071 - Life cycle assessment - Critical review processes and reviewer competencies: Additional requirements and guidelines to ISO 14044:2006
- ISO 14072 - Life cycle assessment - Requirements and guidelines for organizational life cycle assessment
- ISO 14073 - Water footprint - Illustrative examples on how to apply ISO 14046

LCA

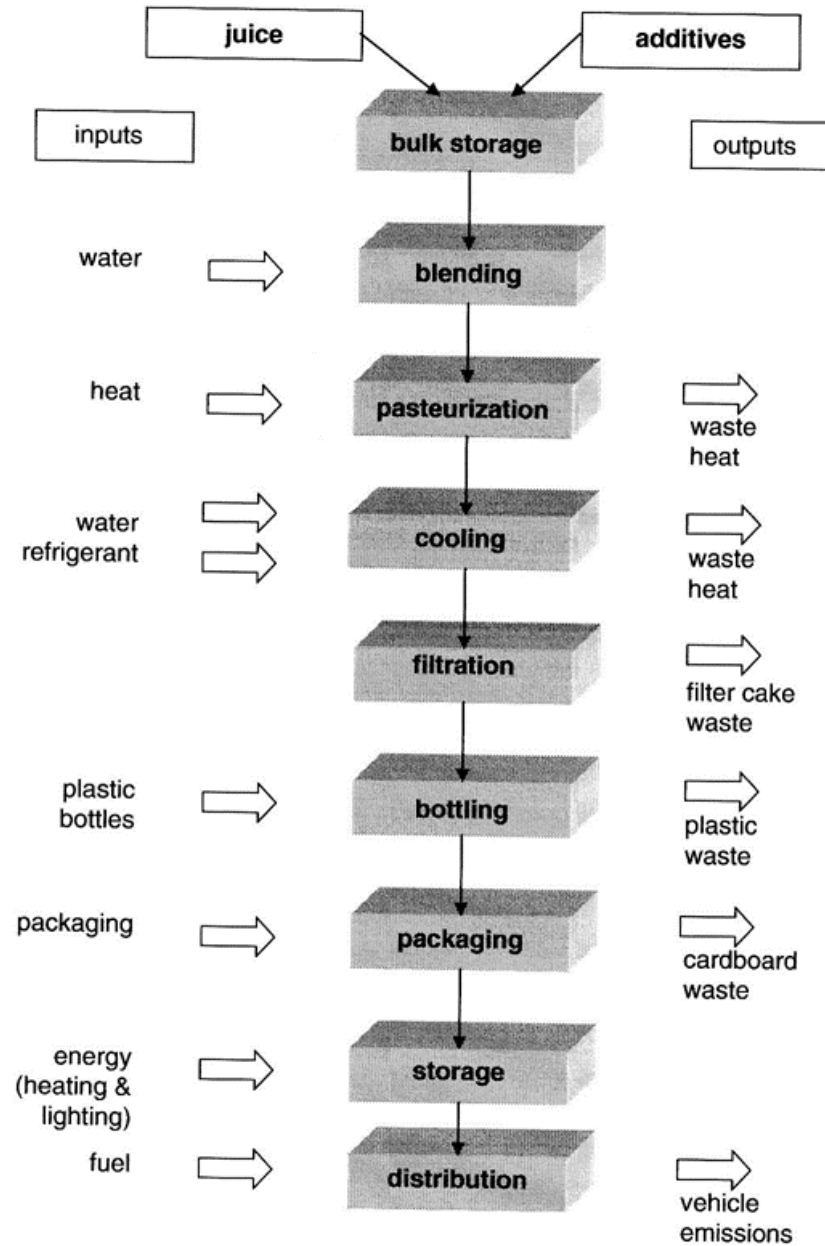
- Choices, assumptions, goal and scope need to be transparent and clearly documented
- Limitations and caveats should be stated
- Data sources should be clearly documented and referenced
- Model used should be clearly documented

LCA Phases - Iterative



LCA goal and scope

- Goal
 - Intended application
 - Reasons for the study
 - Intended audience
- Scope
 - Functions of the system being studied
 - Functional unit
 - Product system being studied
 - Product system boundaries
 - Allocation procedures



Ref: ISO 14001:
Environmental Systems
Handbook

LCA goal and scope

- Scope
 - Types of impact
 - Method of impact assessment
 - Interpretation
 - Data requirements
 - Assumptions
 - Limitations
 - Data quality requirements
 - Critical review
 - How it is reported and format of the report

Function and Functional Unit

- E.g.
 - Product – ton of product
 - Highways
 - Road – km of road
 - Road use – km travelled
 - Subdivision, housing - land – hectare or m^2 of land
 - Water treatment - water – m^3 of water
 - Excavation equipment
 - Tons or m^3 of earth moved
 - TV
 - Product – per screen
 - Use – hours used

Inventory analysis/impact assessment

- Break down into processes involved in manufacture and/or use, disposal
 - Gets large!
 - Decide which processes you want to examine – system boundaries
- Quantify material and energy inputs and outputs for each process
- Allocating an impact to each of those material and energy inputs and outputs
 - For material coming from outside the system boundary, impacts may be unquantified – system expansion
 - Multiple products from the same process
 - Weight impact – mass basis or economic basis

Impact data for materials/processes

- soca
- EuGeos' 15804-IA
- NEEDS
- ecoinvent
- PSILCA
- ESU World Food
- GaBi
- ELCD
- LC-Inventories
- Social Hotspots
- ProBas
- bioenergiedat
- Agribalyse
- USDA
- Ökobaudat
- Agri-footprint
- Comprehensive Environmental Data Archive (CEDA)

Impact data for materials/processes

- Journal papers, technical reports
- Data may be specific for region, country, process
- Calculations

Impact assessment

- Selection of impact categories
- Tally up each impact type to give an overall impact
 - E.g CO₂ per km of road

Interpretation

- Discussion/implications
 - What do the results mean in terms of the goal and scope of the study?
 - Any issues
 - Data completeness, sensitivities, uncertainties

Types of LCA

- Attributional – Looking at the impact of one system only (what we are doing)
- Consequential – Comparing effects of decisions made
 - Choosing between two or three methods of asphalt manufacture
 - Choosing between using recycled concrete of new concrete
- Process LCA – (what we are doing)
- Economic Input Output LCA
 - Aggregates the processes into commodities
 - Environmental impacts of each economic activity
- Hybrid Process LCA, EIO LCA
- Dynamic LCA – includes time dependent, temporal impacts